UNIVERSITY OF NAPLES FEDERICO II

Hardware and Software for Big Data

PROJECT REPORT

Multiclass Sentiment Classification using Kafka Streaming

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Abstract

This project presents a real-time multiclass sentiment classification system built using Apache Spark, Apache Kafka, and Spark NLP. Tweets are streamed through Kafka, transformed into semantic embeddings using the Universal Sentence Encoder, and classified into four sentiment classes: positive, negative, litigious, and uncertainty. The system is designed to process streaming data efficiently while maintaining reliable predictive performance. Experimental evaluation demonstrates an average accuracy of approximately 69%, confirming the feasibility of integrating distributed machine learning with real-time stream processing.

Keywords

Apache Spark, Apache Kafka, Spark NLP, Machine Learning, Sentiment Analysis, Streaming Analytics, Logistic Regression, Natural Language Processing

1. Challenge Description

1.1 Problem Statement

The objective of this project is to develop a scalable real-time sentiment analysis system capable of classifying streaming tweets into four sentiment categories. The solution combines distributed processing with machine learning to support low-latency predictions on continuous data streams.

1.2 Dataset

Dataset: Sentiment Dataset with 1 Million Tweets (Kaggle)
Training Samples: 5,000
Classes: Positive, Negative, Litigious, Uncertainty
Format: CSV containing tweet text and sentiment labels.

2. Proposed Methodology and System Architecture

2.1 System Architecture

The proposed solution follows a streaming architecture consisting of three main components: a Kafka Producer, an Apache Kafka broker, and a Spark Consumer. The producer continuously publishes tweets to a Kafka topic, while the consumer processes each micro-batch using Spark Structured Streaming and performs sentiment prediction in real time.

2.2 Machine Learning Pipeline

Stage	Description
DocumentAssembler	Converts raw tweet text into Spark NLP document format.
Universal Sentence Encoder	Generates 512-dimensional semantic embeddings.
EmbeddingsFinisher	Transforms embeddings into ML-ready vectors.
SQLTransformer	Extracts feature vectors.
StringIndexer	Encodes sentiment labels numerically.
Logistic Regression	Performs multiclass sentiment classification.
IndexToString	Converts predictions back to readable labels.

2.3 Key Technical Features

• Apache Spark for distributed processing
• Apache Kafka for real-time data streaming
• Spark NLP for semantic feature extraction
• Logistic Regression classifier

• Fault-tolerant checkpointing
• Continuous streaming inference with batch evaluation

3. Experimental Setup and Results

3.1 Experimental Configuration

Parameter	Value
Training Set Size	5,000 samples
Batches Processed	438
Spark Version	3.5.0
Spark NLP	5.5.3
Driver Memory	4 GB
Kafka Partitions	1

3.2 Overall Performance

Key Metrics
• Mean Accuracy: 69.00%
• Mean Precision: 72.79%
• Mean Recall: 69.00%
• Peak Accuracy: 100%
• Final Cumulative Accuracy: 69.00%

3.3 Discussion

The experimental evaluation demonstrates that the proposed streaming architecture maintains stable predictive performance over 438 consecutive micro-batches. The use of semantic sentence embeddings significantly improves text representation, while Spark Structured Streaming ensures efficient real-time inference with minimal latency.

4. Performance Analysis and Conclusion

4.1 Performance Analysis

The proposed system successfully processed 438 streaming micro-batches without interruption, demonstrating the robustness of the Apache Kafka and Apache Spark integration. The Universal Sentence Encoder generated high-quality semantic embeddings, enabling the Logistic Regression classifier to achieve stable predictive performance across four sentiment classes.

Although the model achieved a final cumulative accuracy of 69%, the precision remained higher than recall, indicating that the classifier produced relatively few false-positive predictions. The observed performance fluctuations are primarily attributed to variations in batch size and class distribution during streaming.

4.2 Conclusion

This project demonstrates that combining Apache Spark Structured Streaming, Apache Kafka, and Spark NLP provides an effective framework for real-time multiclass sentiment analysis. The system satisfies all project requirements while maintaining reliable streaming performance and low processing latency. The developed pipeline can be extended to larger datasets and production-scale streaming environments with minimal architectural changes.

5. Future Work

Several improvements can further enhance the proposed sentiment classification system:

• Replace Logistic Regression with transformer-based models such as BERT, RoBERTa, or DistilBERT.
• Perform systematic hyperparameter optimization to improve classification performance.
• Address class imbalance using advanced resampling techniques.
• Deploy the pipeline using Docker and Kubernetes for scalable cloud-based streaming.
• Integrate a real-time dashboard for monitoring predictions and performance metrics.

GitHub Repository

Source Code:
https://github.com/subhadip191

References (IEEE Style)

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